PART 1. INTRODUCTION

1. Problem Statement

IT recruitment requires candidates and recruiters to compare many details, including technical skills, experience, language, location, and working conditions. Candidates must identify suitable vacancies, while recruiters must review CVs and decide which applicants deserve closer attention. As the number of jobs and profiles grows, doing this manually becomes slow and inconsistent [10].

Most job portals support posting, searching, and applying, but they do not fully explain why a job fits a candidate or why one applicant ranks above another. CareerFit therefore treats CV–JD matching and personalized recommendation as related but separate tasks. The first compares a specific CV and job; the second prioritizes jobs from a candidate's broader profile and preferences.

Automation can reduce repetitive work, but a similarity score should not directly trigger an important action. User consent, application state, previous interactions, thresholds, quotas, and special conditions also matter. CareerFit uses a Human-in-the-Loop design so users can review important actions and administrators can trace what happened.

This thesis develops CareerFit IT AutoPilot, an IT-focused web platform that combines job discovery, reviewed CV and JD processing, matching, recommendation, Rocchio feedback, AutoFit policies, secure email actions, recruiter talent discovery, and audit records. The system helps users make decisions but does not replace recruitment judgment.

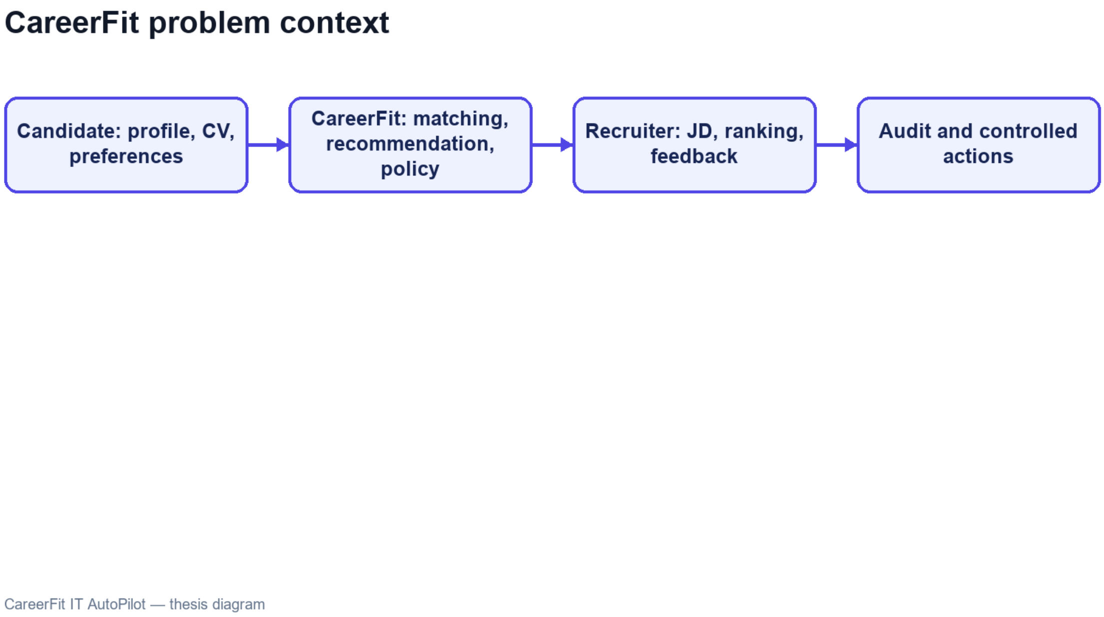

Figure 1.1. CareerFit problem context and principal information flows

2. Background and Related Work

The first motivation is to reduce fragmentation. Candidates often move between job search, CV management, application history, and email, while recruiters switch between vacancy management, applicant review, candidate discovery, and reporting. CareerFit brings these activities into one role-based system and uses email as an additional action channel.

The second motivation is explainability. TF-IDF and cosine similarity do not capture meaning as deeply as modern embedding models, but their terms and weights can be inspected. This makes them suitable for an academic baseline in which the scoring process, limitations, and effect of feedback must be demonstrated clearly.

The third motivation is controlled adaptation and automation. AutoFit places policy checks between a score and an action, considering consent, thresholds, previous interactions, cooldown, quota, time zone, and quiet hours. Rocchio feedback then provides a transparent way to adjust later rankings from explicit positive and negative judgments without overwriting the original representation.

Related work includes classical lexical retrieval, learned resume-job representations, explainable recommendation, and human-centered governance. Lexical methods provide inspectable weights and matching reasons, while recent dense approaches improve semantic representation but require stronger data and evaluation assumptions [6]-[8]. Research on algorithmic hiring and explainable recommendation also shows that model performance, fairness, user control, and explanation quality must be evaluated separately [9], [10], [14]. CareerFit addresses a narrower integration gap by combining portal workflows, explainable scoring, feedback learning, policy-controlled actions, and audit records in one academic prototype.

3. Research Objectives

3.1 Overall Objective

The overall objective of this thesis is to design, implement, and evaluate a web-based Human-in-the-Loop recruitment automation platform for the IT domain. The platform is intended to support CV–job description evaluation, personalized job recommendation, feedback-based adaptation, and policy-driven actions while preserving user control, explainability, security, and auditability.

3.2 Specific Objectives

To achieve the overall objective, the thesis defines the following specific objectives:

- Design a role-based web platform for guests, candidates, recruiters, and administrators, with protected workflows and a content-reporting path for suspicious Jobs and CVs.
- Provide candidate profile and CV management, including upload, manual drafts, OCR-assisted extraction, section review, user correction, confirmation, validation, and quality feedback before matching.
- Implement separate CV–JD matching and candidate-profile recommendation workflows using a shared text-normalization, TF-IDF, and cosine-similarity pipeline.
- Present normalized scores, relevance labels, matching reasons, validation signals, and a separate Potential explanation based on transferable skills, skill families, shared foundations, and seniority compatibility.
- Integrate explicit user feedback and apply the Rocchio relevance-feedback method to update learned representations and recompute rankings reproducibly.
- Implement AutoFit policies that convert matching results into controlled outcomes according to consent, thresholds, interaction state, quota, cooldown, time-zone, and quiet-hour rules.
- Support expiring email-action links with hashed token storage, a non-mutating confirmation page, POST execution, action-state tracking, and audit logging.
- Evaluate the algorithmic behavior and system workflows using reproducible experiments, automated tests, scripted acceptance scenarios, and security and reliability checks appropriate to the project scope.

4. Research Subjects and Scope

4.1 Research Subject

The research subject is the IT recruitment workflow supported by a web platform, including CV and Job-description text, lexical ranking and recommendation, explicit feedback signals, policy-controlled automation, and auditable user and system actions.

4.2 Research Scope

This thesis focuses on an academic prototype for IT recruitment. Its functional scope includes a public job portal; authenticated Candidate, Recruiter, and Administrator workspaces; reviewed CV and profile management; recruiter company profiles; Job drafts and publishing; CV–JD matching; personalized recommendation; applications; Talent Pool bookmarks and invitations; explicit feedback; AutoFit policies; notifications; analytics; content reporting and administrator moderation; and audit records.

The data-processing scope includes extraction from supported CV documents, image preprocessing and OCR fallback, OCR cleanup, section review, bilingual-oriented text normalization, TF-IDF vectorization, cosine-similarity scoring, a versioned skill-transfer Potential assessment, relevance labeling, and Rocchio feedback updates. PostgreSQL is the main data store, and Flyway manages database changes. The React and TypeScript frontend calls REST APIs provided by the Spring Boot backend.

The thesis does not attempt to implement a complete enterprise applicant tracking system. Interview scheduling, offer management, payroll, and full employee lifecycle management are outside the core scope. The system does not automatically submit applications to external third-party job boards, and it does not use a large language model to make unrestricted recruitment decisions. Distributed microservices, large-scale message brokers, and semantic embedding models may be discussed as future extensions but must not be presented as implemented core components unless they are added and verified before the final report is submitted.

The evaluation scope is limited. Controlled or synthetic datasets can show whether Rocchio changes rankings in the expected direction, but they cannot prove recruitment quality in a real production setting. The source and purpose of scraped, seeded, or synthetic data must be stated. Claims about usability, performance, security, or production readiness are limited to the test environment, participants, procedures, and evidence available during the final evaluation.

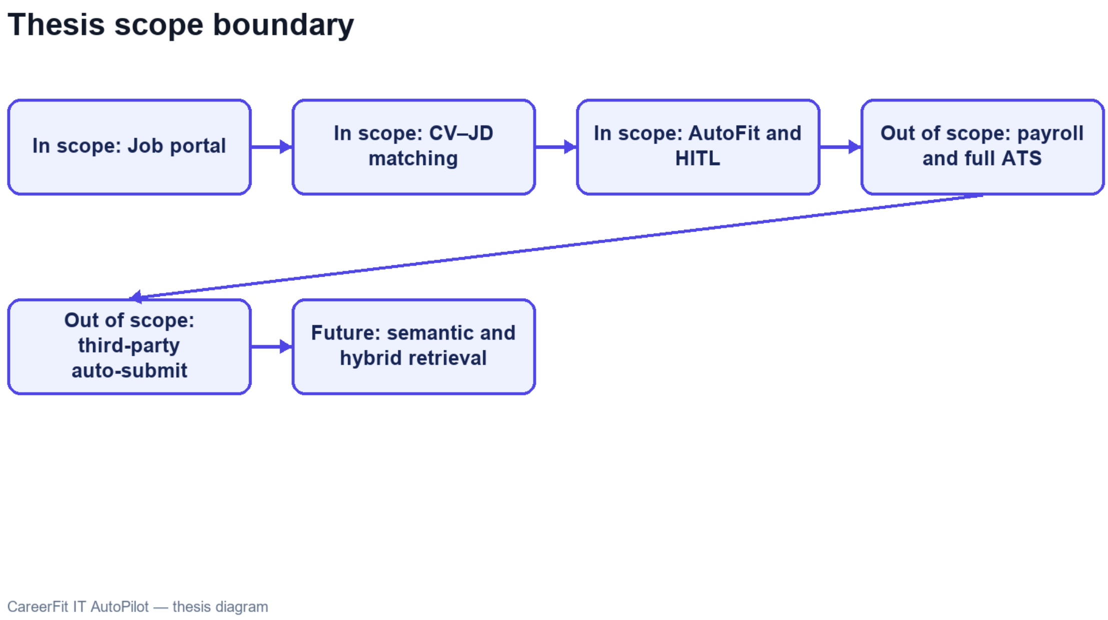

Figure 1.2. Scope boundary of the CareerFit thesis

5. Research Methods and Content

5.1 Research Methods

This thesis follows an applied software-engineering research approach. The work combines requirement analysis, architecture and data design, incremental implementation, a controlled algorithm experiment, automated backend and browser testing, runtime observation, and threats-to-validity analysis. Implementation claims are checked against source code, migrations, API contracts, logs, test reports, and generated evaluation artifacts.

5.2 Research Content

The research models CareerFit as a Perception-Decision-Action-Learning-Audit workflow. It first identifies recruitment roles, requirements, and control boundaries; then designs and implements the data, matching, recommendation, feedback, automation, email action, and audit modules. Finally, the study evaluates algorithm behavior, software correctness, browser workflows, authorization, runtime health, and local latency within the stated experimental scope.

6. Key Contributions

The principal contribution of this thesis is the design and implementation of an integrated, controlled recruitment workflow rather than the invention of a new machine-learning model. The contributions are summarized as follows:

- An integrated recruitment platform that combines public job discovery, reviewed CV processing, CV–JD matching, personalized recommendation, application management, Recruiter Jobs, Talent Pool, and a report-moderation workflow.
- A transparent scoring design that keeps the TF-IDF/cosine direct-match score separate from the versioned Potential assessment and explains transferable skills, shared foundations, and seniority checks.
- A feedback-learning workflow based on the Rocchio method, designed to update learned representations from explicit feedback and to support reproducible recomputation rather than uncontrolled cumulative modification.
- A policy-driven AutoFit layer that separates relevance scoring from business actions and applies Human-in-the-Loop controls, user consent, operational constraints, and previous interaction states.
- An actionable email mechanism with expiring, hashed tokens, a non-mutating GET confirmation page, POST redemption, and action-state tracking.
- A traceable evaluation structure that distinguishes algorithm experiments from functional, integration, acceptance, performance, reliability, and security evidence, while explicitly documenting threats to validity.

Thesis objective	Implemented contribution	Evaluation evidence
Analyze IT recruitment workflows	Role-specific requirements and Human-in-the-Loop boundaries	Use cases and traceability matrix
Implement CV–JD matching	TF-IDF, cosine scoring, explanations, and persisted Matchings	Algorithm benchmark and backend tests
Learn from feedback	Rocchio updates followed by scheduled recomputation	Baseline-versus-learned ranking metrics
Support controlled automation	AutoFit policy, quota, cooldown, notification, and audit controls	Security tests and P0 E2E flows
Deliver an integrated platform	Spring Boot, React/TypeScript, PostgreSQL, and Flyway	Build, runtime health, and browser evidence

Table 1.1. Mapping of objectives, contributions, and evidence

7. Thesis Structure

The report contains an Introduction, four chapters, and a Conclusion. Chapter 1 defines the problem, stakeholders, requirements, and use cases. Chapter 2 presents the theoretical background and solution design. Chapter 3 explains the implementation. Chapter 4 presents the evaluation method, evidence, and threats to validity. The Conclusion discusses achievements, limitations, and future work.